

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer bridging core model training and optimization with production-scale agentic systems for 1M+ active users. Specialized in triplet contrastive loss fine-tuning, LLM SFT (Unsloth), real-time WebSocket speech ASR/TTS streams, and custom Java-based ReAct agent runtimes. Contributor to Spring AI, combining machine learning research with robust engineering to deliver production-grade AI platforms.

Technical Skills

Languages & Backend: Python, Java, SQL, Spring Boot, FastAPI, PostgreSQL, SAP HANA, Data Engineering, C++
AI/ML Engineering: Agentic AI, Model Context Protocol (MCP), SFT, Triplet Contrastive Fine-Tuning, Hard Negative Mining, vLLM
Speech & Audio ML: STFT, Mel-spectrograms, CNNs (AlexNet-EfficientNet), ASR (Whisper, Vosk, NeMo, IndicConformer), TTS (Bark, Coqui)
AI Frameworks: Spring AI, LangGraph, CrewAI, AutoGen, LangChain, Hugging Face Transformers, LlamaIndex, n8n
Platforms & Models: Azure OpenAI, SAP AI Core, Amazon Bedrock, Google Vertex AI, Claude, Gemini, Qwen
Infrastructure & Security: Docker, Kubernetes, Cloud Foundry, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization (BIO NER)

Experience

TalenTeam Ltd., UK (via Bigscal Technologies)

AI/ML Engineer Oct 2024 - Present
Remote

- Deployed a production-scale agentic platform and multi-agent graphs serving 1M+ active ERP users across 120+ SuccessFactors tenants.
- Reduced multi-agent execution overhead by 3x and token consumption by 70% by designing a capability contract layer executing nested capability profiles inside a single ReAct loop.
- Fine-tuned E5-large-v2 via triplet contrastive loss with hard negative mining in PyTorch, and trained Qwen2.5-7B via Unsloth SFT on A100s (served via vLLM on Azure) for domain search and job architecture mapping.
- Developed the Prompt Lab evaluation control plane utilizing a FastAPI + UI + n8n architecture to capture durable reasoning trajectories and human preference annotations for LLM-as-a-Judge evaluations.
- Implemented a hierarchical RAG pipeline indexing paragraph, page, and chapter layers with PostgreSQL/PGVector and signature-based content deduplication for 10,000+ patent documents, achieving 90%+ Recall@10.
- Secured agentic workflows against OWASP LLM Top 10 via observer-based tool monitoring and fine-tuned SpaCy token-classification models for multilingual BIO Named Entity Recognition (NER) to automate PII redaction.

Healthray (Bigscal Technologies)

AI/ML Engineer Intern Apr 2024 - Sep 2024
Surat, India

- Developed asynchronous FastAPI WebSocket streaming audio pipelines for Indic language ASR (Whisper, Vosk, NeMo) and built quantized Llama 3 clinical summarization POCs on 116K+ records.
- Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Open Source & Projects

Spring AI (spring-projects/spring-ai) | Open Source

Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.
Java, Spring Boot

Modular Agent Harness & Memory System ("Brain") | Open Source

Architected an open-source plug-and-play agentic framework designed as modular, composable building blocks for flexible enterprise workflows, implementing a formal Java-based ReAct loop (Thought-Action-Observation) with step budgets, tactical pgvector tool routing, and dynamic MCP settings caching.
Java, Spring Boot, PostgreSQL, WebSockets

AI Agent Security Red Teaming - OpenAI | Security Researcher

Designed adversarial prompt-injection pipelines targeting guardrails to evaluate security boundaries across 2,000+ agent test cases.
Python, Kaggle

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University 2022 - 2024

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900)	Microsoft
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